Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State the no-air-resistance assumption.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A toy model predicts position (-40, 60) km with bounds ±4 km in x and ±6 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_f = v_i + g\Delta t$; $\Delta y = v_i\,\Delta t + \frac{1}{2}g(\Delta t)^2$.
Downward positive for these equations; $g = 10$ m/s². Ideal free fall: gravity is the only force.

Worked reasoning: A ball dropped from rest has downward speeds 0, 10, 20 m/s at 0, 1, 2 s in ideal free fall with g=10 m/s². Equal times give equal velocity changes.

Your turn: Sketch its velocity-time graph with downward positive. Explain why the positions are not equally spaced.

Marzano check - circle: ① ② ③

03 Represent and interpret

Two balls of different mass are released together in a vacuum from the same height. Compare acceleration and arrival time.

Marzano check - circle: ① ② ③

04 Solve and justify

A ball is thrown upward at 20 m/s. Use upward positive. Find time to the top and velocity at 3 s.

Marzano check - circle: ① ② ③

05 Analyze the reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the claim.

Marzano check - circle: 1 2 3

06 Choose a method

An a-versus-F_{net} graph has slope 2 (m/s²)/N. Find mass and acceleration at 3 N. Explain why a driving-force plot can have an offset.

Marzano check - circle: 1 2 3

07 Exit check

A ball starts with downward velocity 5 m/s and falls for 2 s. Find final speed and displacement.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A ball is thrown up at 15 m/s from a 20 m platform. Use g=10 m/s². Find its height at 2 s and its time to reach the ground.

Marzano check - circle: 1 2 3